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Comparative Evaluation of Performance Trade-Off Under Permanent Magnet Volume Reduction in SPM, IPM, and PM-Assisted SynRM

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Abstract

Rising costs, supply uncertainty, and geopolitical dependence associated with rare-earth permanent magnets have created an urgent need to reduce permanent magnet (PM) usage in electric machine designs while maintaining high electromagnetic performance. This paper presents a comprehensive comparative investigation of PM reduction strategies for three widely adopted traction machine topologies: Surface Permanent Magnet (SPM), Interior Permanent Magnet (IPM), and Permanent Magnet-Assisted Synchronous Reluctance (PMASynRM) machines. We developed a unified evaluation framework by integrating each machine's electromagnetic characteristics, electrical operating constraints, and practical inverter limitations across a wide operating speed range. Finite element analysis (FEA) is employed to systematically investigate the effects of progressive PM volume reduction by independently varying magnet width and length while keeping stator geometry, winding configuration, material properties, and operating conditions. We reduced PM dimensions from 100% to 30% of the baseline design to evaluate their impact on torque-speed characteristics, efficiency, and flux-weakening capability. The results reveal that the impact of PM reduction depends strongly on both machine topology and the magnet reduction strategy. Across all machine topologies, reducing magnet width consistently preserves higher torque capability, efficiency, and high-speed operating performance compared with reducing magnet length. Among the machines investigated, the SPM topology exhibits the greatest sensitivity to PM reduction, particularly in the constant-torque region, owing to its complete reliance on permanent magnet excitation. In contrast, the IPM and PMASynRM machines show much greater tolerance to reduced PM volume because the additional contribution of reluctance torque from rotor saliency improves torque retention, extends the constant-power speed range, and reduces efficiency degradation. These findings provide valuable design insights into the trade-offs between permanent magnet utilization and machine performance, offering practical guidelines for minimizing rare-earth material consumption while maintaining competitive traction performance in next-generation electric vehicles and other sustainable electrification applications.

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1. Introduction

Rare-earth permanent magnets remain central to achieving high torque density and efficiency in many electric drive systems, yet their supply chain faces concentration risk and policy-driven disruption. This exposure is increasingly relevant to traction applications, where high production volumes amplify the impact of price volatility and availability constraints on total drivetrain cost and design choices [1]. In parallel, the operating requirements of electrified transport continue to push traction machines towards wider speed ranges and sustained high-speed operation. In addition, the inverter's voltage margin and the associated flux-weakening become design-limiting, and the permanent-magnet flux strongly affects the achievable speed range and efficiency [2].

A substantial body of work has established operating-limit concepts for synchronous motor drives in flux weakening and has clarified how current and voltage constraints shape the feasible torque-speed envelope [3]. However, published studies often focus on control strategy development or optimizing a single machine family, rather than quantifying how magnet-volume reduction alters performance across fundamentally different rotor topologies under a common set of constraints.

Comparative studies across motor technologies exist, but comprehensive comparisons remain relatively rare in the literature, and they seldom provide a geometry-resolved sensitivity view of permanent-magnet reduction [4]. In addition, prior work examining magnet geometry often focuses on waveform quality, torque ripple, or back-EMF harmonics for specific machine topology, but is not typically framed as a systematic, topology-spanning sensitivity study of magnet thickness and width with explicit trade-off metrics. Furthermore, researchers usually treat the behavior of "zero-PM" operation implicitly by switching to a different machine class (for example, a synchronous reluctance machine) rather than benchmarking it as an intentional operating point for PM-derived topologies. This leaves limited guidance on residual capability, efficiency penalties, and control implications when the magnet contribution approaches zero [5]. Taken together, four key gaps motivate this work: (i) limited studies comparing magnet reduction across different machine topologies, (ii) insufficient distinction between reductions in magnet width and thickness, (iii) limited evaluation of progress toward zero-magnet operation in PM-based machines, and (iv) inadequate analysis of the high-speed performance trade-off associated with reduced magnet volume and increased current-related losses.

Several studies have investigated strategies for reducing rare-earth permanent-magnet usage while maintaining acceptable electromagnetic performance. Kano et al. [6] reviewed rare-earth-reduced and rare-earth-free PMSM technologies, including ferrite-based IPMSMs and PM-assisted synchronous reluctance machines, highlighting the increasing emphasis on reducing dependence on high-energy rare-earth magnets. Guglielmi et al. [7] specifically investigated permanent-magnet minimization in PM-assisted synchronous reluctance motors and demonstrated that magnet volume can be significantly reduced through appropriate rotor design while maintaining machine performance; however, they also emphasized that decreasing magnet dimensions affects the magnet operating point and demagnetization capability. More recently, Khonongbua et al. [8] developed a hybrid-magnet multi-flux-barrier IPM motor in which partial substitution of rare-earth magnets with ferrite, combined with rotor optimization, reduced rare-earth usage while maintaining electromagnetic torque and efficiency close to those of the

benchmark design. These studies demonstrate that PM reduction is feasible, but its performance impact depends strongly on magnet configuration, rotor geometry, and the resulting magnetic circuit.

PMASynRM has received particular attention as a magnet-reduction strategy because a significant portion of its electromagnetic torque can be produced through rotor saliency rather than PM excitation alone. Jani and Jamnani [9] compared PMASynRM designs employing ferrite and rare-earth magnet materials and investigated the trade-off among magnet quantity, torque, efficiency, and torque ripple. Recent work has further demonstrated the potential of hybrid-magnet PMASynRM designs for traction applications. Kumar et al. [10], for example, reported optimized high-speed PMASynRM configurations achieving substantial reductions in rare-earth magnet usage while maintaining the required electromagnetic performance. Broader assessments of rare-earth reduction in traction machines likewise identify PMASynRM as a promising magnet-lean alternative because reluctance torque can compensate, in part, for reduced PM excitation. Nevertheless, much of the existing literature concentrates on magnet-material substitution, rotor optimization, or PM reduction within an individual machine topology. A systematic comparison of how independent reductions in magnet width and magnet length influence SPM, IPM, and PMASynRM machines under consistent geometric and electrical operating constraints remains comparatively limited.

As such, this paper addresses the following fundamental question: How does progressively reducing permanent-magnet volume, implemented through controlled changes in magnet width and length, affect the performance trade-offs of representative surface permanent-magnet (SPM), interior permanent-magnet (IPM), and permanent-magnet-assisted synchronous reluctance (PM-assisted SynRM) machines, from low speed to ultra-high speed, including the limiting case of zero permanent-magnet volume. To ensure a fair comparison, identical permanent-magnet volumes are considered across the three machine topologies at each reduction level, and the corresponding electromagnetic, electrical, and operating constraints are consistently applied.

The study therefore provides a systematic basis for quantifying how machine topology and permanent magnet utilization jointly affect torque capability, efficiency, flux-weakening capability, and high-speed performance. The results are intended to provide practical design guidelines for traction motors by clarifying the trade-offs among machine topology, electromagnetic performance, efficiency, and permanent magnet material utilization, thereby supporting the development of high-performance traction machines with reduced reliance on rare-earth permanent magnets.

The main contribution of this study is a unified FEA-based comparative framework for systematically evaluating permanent-magnet reduction across different synchronous machine topologies. In contrast to evaluating PM reduction within a single topology, the proposed methodology applies consistent geometric, material, and operating constraints to SPM, IPM, and PMASynRM machines and independently varies magnet width and magnet length at 100%, 80%, 60%, 40%, and 30% of their baseline dimensions. This controlled approach enables the influence of PM-reduction direction to be distinguished from topology-dependent torque-production mechanisms and allows the sensitivity of each machine to reduced PM usage to be directly compared. An equal-PM comparison is further employed to assess how rotor topology and magnet utilization influence torque-speed capability and efficiency when the same amount of PM material is used.

2. Evaluation of Machine and Magnet Topologies

In traditional radial-flux configurations of three-phase Permanent Magnet Synchronous Motors (PMSMs), the magnetic flux flows radially across the air gap between the centrally located rotor and the surrounding stator. This includes both the magnetic flux

produced by the three-phase stator windings and the flux generated by the rotor-mounted permanent magnets. The stator magnetic field rotates synchronously with the rotor, so the stator and rotor magnetic fields appear stationary relative to each other in the rotating reference frame.

Because the stator magnetic flux crosses the air gap radially, the rotor magnetic flux must also be oriented radially to ensure effective electromagnetic coupling. A commonly used analytical framework for describing this interaction is the d-q axis representation. In PMSMs, the stator d-axis is aligned with the rotor magnetic field axis, so the stator d-axis flux coincides with the rotor d-axis flux.

Typical PMSM topologies are primarily classified by the configuration and placement of the permanent magnets. These include Surface Permanent Magnet (SPM), Interior Permanent Magnet (IPM), and Permanent Magnet Assisted Synchronous Reluctance (PMASynRM). Each topology exhibits electromagnetic characteristics, differing torque production mechanisms, and performance trade-offs in efficiency, power density, cost, and control complexity. This section presents a detailed evaluation of these PMSM topologies to highlight their operational principles and comparative foundations.

2.1. SPM Topology

In SPM machines, as shown in Figure 1, the permanent magnets are mounted directly on the rotor surface and are typically radially magnetized, so their magnetization direction aligns with the rotor d-axis and coincides with the stator d-axis flux. Consequently, the magnetic flux produced by the magnets is primarily directed along the d-axis, establishing a dominant radial flux path across the air gap. As a result, the fundamental air-gap magnetic field is predominantly aligned with the rotor d-axis, with relatively low spatial distortion of the air-gap flux-density waveform [6].

From an electromagnetic field perspective, the air-gap flux density can be decomposed into radial (B_{ra}) and tangential (B_{ta}) components in cylindrical coordinates. For SPM machines, analytical and experimental studies have shown that the air-gap field is predominantly governed by the radial component B_{ra}, while the tangential component B_t remains minimal under ideal conditions and primarily arises from slotting effects and higher-order harmonics. This behavior is symmetric rotor structure, which suppresses reluctance variation along the q-axis and limits cross-axis coupling effects. As a result, the fundamental air-gap field is largely aligned with the d-axis and shows minimal distortion [11].

Furthermore, due to the non-salient rotor geometry, the d- and q-axis inductances are approximately equal $(L_d \approx L_q)$, indicating that torque production in SPM machines is almost entirely due to the interaction between the stator current and the permanent magnet flux, with negligible contribution from reluctance torque [12].

From a magnet topology perspective, the SPM machine has a relatively simple rotor structure because the magnets are mounted directly on the rotor surface, making it easier to design and manufacture. Furthermore, the uniform air gap and negligible rotor saliency (due to $L_d \approx L_q$) result in a nearly sinusoidal air-gap flux density, enabling smooth electromagnetic torque production with low torque ripple that minimizes harmonic and mechanical losses. However, surface-mounted magnets require retaining sleeves for mechanical integrity and limit maximum rotor speed due to centrifugal forces.

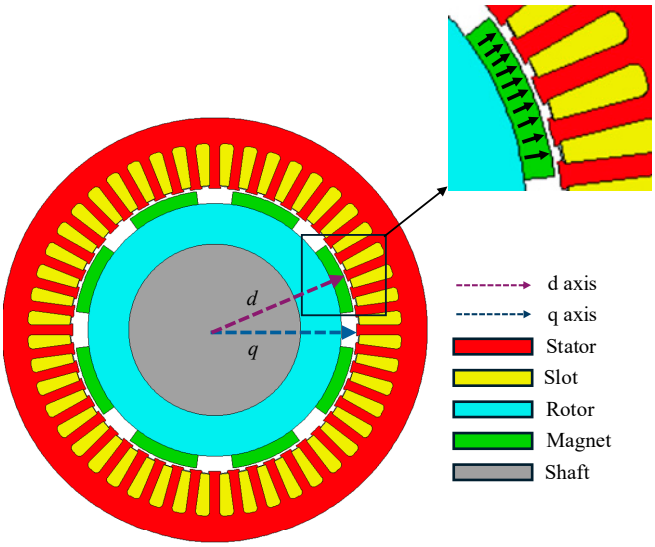


Figure 1. SPM topology showing dq axis and PM magnetization direction (inset).

2.2. IPM Topology

In IPM machines, as shown in Figure 2, the magnets are embedded within the rotor iron and are commonly arranged in a V-shaped configuration [9]. In such structures, the individual magnet segments are intentionally inclined relative to the radial direction and, therefore, are not locally aligned with the rotor d-axis. Despite this, the rotor geometry is symmetric about a central magnetic axis. As a result, the magnetic flux produced by the individual magnet segments combines to form a resultant flux aligned with this symmetry axis, which is defined as the machine's d-axis. Hence, although the local magnetization directions deviate from the d-axis, the rotor's effective magnetization remains d-axis oriented and coincides with the stator's d-axis flux.

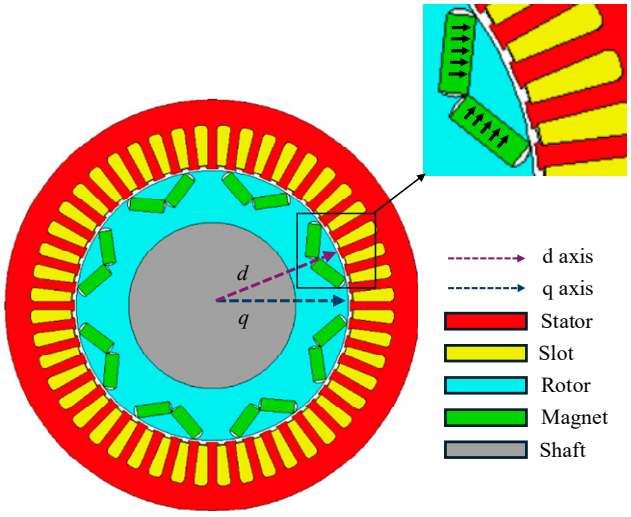


Figure 2. IPM topology showing dq axis and PM magnetization direction (inset).

Consequently, the magnetic flux generated by the magnets is primarily directed along the d-axis. As the flux propagates from the embedded magnets toward the air gap, it passes through regions of varying magnetic permeability, including iron bridges. These structural features introduce anisotropic magnetic reluctance in the rotor, resulting in a clear distinction between the d-axis and q-axis inductances $(L_d \neq L_q)$ [13–15].

Therefore, the flux does not remain purely radial in the air gap. Instead, the distortion of flux paths caused by rotor saliency results in both radial (B_{ra}) and tangential (B_{ta})

components of the air-gap flux density [16]. The tangential component is particularly significant because it contributes to reluctance torque in addition to the magnetic torque produced by the permanent magnets [17].

From a magnetic topology perspective, the IPM machine features a more complex rotor structure because the permanent magnets are embedded within the rotor core rather than mounted on the surface. This enhances mechanical integrity and enables high-speed operation. Furthermore, rotor saliency ($L_d \neq L_q$) produces a non-uniform air gap, leading to a distorted air-gap flux density with both radial and tangential components. This facilitates the combined generation of magnetic and reluctance torque, thereby improving torque density and efficiency.

In addition, the embedded magnet configuration provides effective flux-weakening capability, allowing operation over a wide constant power speed range [18,19]. However, the embedded topology increases rotor design complexity and manufacturing cost, and the resulting non-uniform flux distribution introduces harmonic distortion, which can lead to torque ripple, acoustic noise, and vibration, thereby requiring more advanced control strategies to achieve optimal performance [19].

2.3. PMASynRM Topology

The PMASynRM, as shown in Figure 3, combines a flux-barrier rotor structure with embedded permanent magnets, enabling a hybrid torque production mechanism that integrates both magnetic and reluctance effects. In this topology, the magnets are strategically placed within selected flux barriers so that their magnetization reinforces the d-axis flux path while preserving the inherent reluctance-based rotor. The magnetization direction is intentionally aligned with the low-reluctance axis (d-axis). At the same time, multiple flux barriers introduce strong magnetic anisotropy, producing a pronounced difference between d- and q-axis inductances $\left(L_d \neq L_q\right)$. These barriers act as high-reluctance regions along the q-axis, effectively restricting flux flow and forcing the magnetic field to concentrate along preferred paths in the rotor [20].

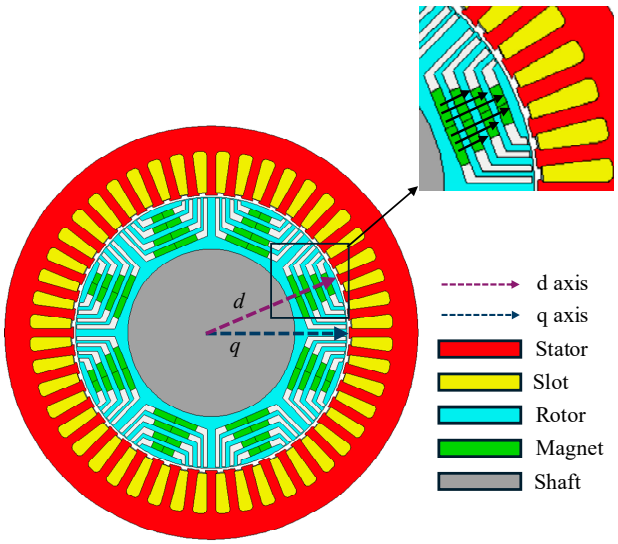


Figure 3. PMASynRM topology showing dq axis and PM magnetization direction (inset).

As a result, the air-gap magnetic field is governed not only by the magnetization direction but also by the rotor's geometry-induced anisotropy. The embedded magnets promote a dominant radial flux component (B_{ra}). The constrained flux paths and permeability transitions caused by the barriers distort the field distribution, leading to the presence of significant tangential components (B_{ta}) in addition to the radial component [21,22].

From a magnet topology perspective, PMASynRMs feature a hybrid rotor structure in which permanent magnets are embedded within flux barriers, enabling a design that prioritizes reluctance torque while using magnets primarily for flux assistance. Furthermore, the pronounced rotor saliency ($L_d \neq L_q$) and the presence of magnet-assisted flux paths create a non-uniform air gap and distorted air-gap flux density, resulting in radial and tangential flux components that enable combined reluctance and magnetic torque while reducing magnet dependency [23–26]. However, integrating magnets within flux barriers increases rotor design complexity and requires careful optimization of barrier geometry and magnet placement. The resulting non-uniform flux distribution can introduce harmonic distortion, leading to torque ripple, acoustic noise, and vibration, which necessitates advanced control strategies to achieve optimal performance [27-32].

Furthermore, the anisotropic rotor design ($L_d \neq L_q$) produces a non-uniform air gap and a distorted air-gap flux distribution, enabling torque generation purely through reluctance effects [33]. However, despite these advantages, SynRMs have lower torque density than permanent-magnet machines, and their pronounced saliency can introduce significant torque ripple, vibration, and acoustic noise. In addition, the relatively low power factor remains a key limitation, often requiring higher inverter volt-ampere ratings and control strategies [34–37].

The investigated machines were evaluated under identical electrical operating constraints to ensure a consistent comparison among the different rotor topologies. A DC-link voltage of 400 V and a peak stator current limit of 480 A were applied to all machines, with MTPA (maximum torque per ampere) (including flux weakening and MTPV (maximum torque per voltage)) [33] control employed for torque production.

In practical applications, however, the impact of structural differences among SPM, IPM, PMSynRM, and SynRM machines on torque capability, magnet utilization, and saliency exploitation is highly complex. Therefore, we used the ANSYS FEA 2025.1.1 tool to model and analyze the effects of permanent magnet configurations across different machine topologies. To ensure a fair comparison, key design parameters, including slot number, pole number, winding configuration, and rotor dimensions, are kept identical across all machine topologies, as summarized in Table 1. This approach ensures that any observed performance variations arise solely from differences in rotor topology and permanent magnet volume. Section 3 presents the mathematical modeling and theoretical evaluation of these machines.

Table 1. Design Specifications of the SPM, IPM, and PMASynRM Machines.

Parameter	SPM	IPM	PMASyncRM
Slot Number	48	48	48
Stator Lam Dia (mm)	198	198	198
Stator Bore (mm)	132	132	132
Pole Number	8	8	8
Rotor Type	Surface Radial	Interior V-shape	PM-assisted SynRM
Rotor Diameter (mm)	130	130	130
Airgap (mm)	1	1	1
Shaft Diameter (mm)	80	80	80
Winding Type	Lap	Lap	Lap
Turns/Coil	6	6	6
Parallel Paths	2	2	2
Phases	3	3	3
PM weight	1.75 Kg	1.75 Kg	1.75 Kg
DC-link voltage (V)	400	400	400
Peak Stator Current (A)	480	480	480
Base Speed (rpm)	4450	3950	4300

Base Electromagnetic Torque (N·m)	296.15	317.74	257.28
Base Electromagnetic Power (kW)	138	131.4	115.9
D-axis Inductance (mH)	0.13	0.16	0.16
Q-axis Inductance (mH)	0.16	0.35	0.41
Efficiency at Base Speed (%)	95.88	97.08	95.78
Control Strategy	MTPA	MTPA	MTPA

3. Electrical, Flux Linkage, and Torque Relations Under Different Machine Topologies

The electrical, flux linkage, and torque relations are fundamental equations for evaluating the performance of permanent magnet synchronous machines (PMSMs). Figure 4 illustrates the overall configuration of a PMSM drive system. On the electrical side, DC power is supplied to the three-phase stator windings via a power inverter. Within the machine, the air-gap flux linkage is established by the interaction between the rotating stator magnetic field generated by the three-phase stator windings and the magnetic field produced by the permanent magnets. On the mechanical side, the interaction between the air-gap flux linkage and the stator current generates electromagnetic torque, which is delivered to a mechanical load, such as an electric vehicle traction system.

Inside the machine, the general form of the flux linkage equations includes contributions from the stator rotating magnetic field and the rotor permanent magnets, and is given, in the dq reference frame, by

$$\begin{bmatrix} \lambda_d \\ \lambda_q \end{bmatrix} = \begin{bmatrix} L_d & L_{dq} \\ L_{qd} & L_q \end{bmatrix} \begin{bmatrix} i_d \\ i_q \end{bmatrix} + \begin{bmatrix} \lambda_{PM} \\ 0 \end{bmatrix} \tag{1}$$

where λ_d and λ_q are the stator flux linkages, L_d and L_q denote the self-inductances of the equivalent stator d- and q-axis windings, and L_{dq} and L_{qd} represent the mutual inductances between the d-axis and q-axis windings. The terms i_d and i_q correspond to the stator d- and q-axis current components, and λ_{PM} is the flux linkage contributed by the rotor permanent magnets. For machines without permanent magnets, such as SynRM, $\lambda_{PM} = 0$.

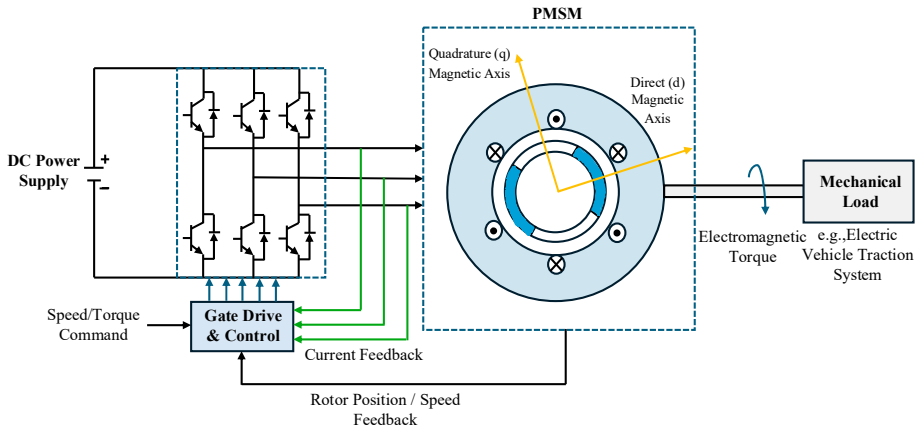


Figure 4. The overall configuration of a PMSM drive system.

When magnetic saturation is considered, the inductances become nonlinear functions of the stator currents, i.e., $L_d = L_d(i_d, i_q)$ and $L_q = L_q(i_d, i_q)$. In addition, the effective permanent magnet flux linkage may vary with the stator currents due to demagnetization effects. Under these conditions, the flux linkage expressions can be written as

$$\begin{bmatrix} \lambda_d \\ \lambda_q \end{bmatrix} = \begin{bmatrix} L_d\left(i_d, i_q\right) & L_{dq}\left(i_d, i_q\right) \\ L_{qd}\left(i_d, i_q\right) & L_q\left(i_d, i_q\right) \end{bmatrix} \begin{bmatrix} i_d \\ i_q \end{bmatrix} + \begin{bmatrix} \lambda_{PM}\left(i_d, i_q\right) \\ 0 \end{bmatrix} \tag{2}$$

The general electrical equations represent the voltage balance among the externally applied stator voltage, the stator winding resistive voltage drop, and the armature reaction-induced voltage components, and are given by:

$$\begin{bmatrix} v_d \\ v_q \end{bmatrix} = R\begin{bmatrix} i_d \\ i_q \end{bmatrix} + \frac{d}{dt}\begin{bmatrix} \lambda_d\left(i_d, i_q\right) \\ \lambda_q\left(i_d, i_q\right) \end{bmatrix} + \omega_e \begin{bmatrix} -\lambda_q\left(i_d, i_q\right) \\ \lambda_d\left(i_d, i_q\right) \end{bmatrix} = R\begin{bmatrix} i_d \\ i_q \end{bmatrix} + \frac{d}{dt}\begin{bmatrix} L_d\left(i_d, i_q\right) i_d \\ L_q\left(i_d, i_q\right) i_q \end{bmatrix} + \omega_e \begin{bmatrix} -\lambda_q\left(i_d, i_q\right) \\ \lambda_d\left(i_d, i_q\right) \end{bmatrix}, \tag{3}$$

where v_d and v_q are the externally applied stator d- and q-axis voltages, and ω_e is the motor electrical speed. The steady-state *dq* voltage is

$$v_d = Ri_d - \omega_e \lambda_q\left(i_d, i_q\right) = Ri_d - \omega_e L_q\left(i_d, i_q\right) i_q \tag{4}$$

$$v_q = Ri_q + \omega_e \lambda_d\left(i_d, i_q\right) = Ri_d + \omega_e\left(L_d i_d\left(i_d, i_q\right) + \lambda_{pm}\right) \tag{5}$$

On the electrical side, both the stator current and stator voltage are subject to operational constraints. The stator current is limited by the rated current capacity of the machine and can be expressed as

$$\sqrt{i_d^2 + i_q^2} = I_{rated} \tag{6}$$

The stator voltage is limited by the inverter's PWM saturation capability. Under the commonly used SVPWM scheme, the voltage constraint can be written as

$$\sqrt{v_d^2 + v_q^2} \approx \sqrt{\left(\omega_e\left(L_d i_d + \lambda_{pm}\right)\right)^2 + \left(\omega_e L_q i_q\right)^2} \le v_{\max} \tag{7}$$

$$\text{or} \quad \sqrt{\left(L_d i_d + \lambda_{pm}\right)^2 + \left(L_q i_q\right)^2} \le \frac{V_{dc}}{\sqrt{2}\omega_e} = \lambda_{\lim,\omega_e} \tag{8}$$

The electromagnetic torque of a PMSM can be expressed as

$$T_{em} = \frac{p}{2}\left[\lambda_d\left(i_d, i_q\right) i_q - \lambda_q\left(i_d, i_q\right) i_d\right] \tag{9}$$

where p is the number of poles. Substituting (2) into (9), the torque expression can be written as

$$T_{em} = \frac{P}{2}\left[\lambda_{PM} i_q + (L_d i_d + L_{dq} i_q) i_q - (L_q i_q + L_{qd} i_d) i_d\right] \tag{10}$$

in which the first term represents the magnetic torque arising from the interaction between the stator q-axis current and the permanent magnet flux, while the remaining terms correspond to the reluctance torque arising from the inductance characteristics of the machine.

While the general flux linkage and electrical relations remain similar across different machine and magnet topologies, variations in magnetic saliency and torque production mechanisms lead to differences in the electromagnetic torque equation.

3.1. SPM Modeling

For an SPM motor, the magnetic circuit is nearly symmetrical along the d- and q-axes, resulting in approximately equal inductances along these two axes $\left(L_d \approx L_q\right)$. Under this condition, the reluctance torque component becomes negligible. Neglecting the

mutual inductance, the electromagnetic torque expression reduces primarily to the permanent-magnet torque component, as follows

$$T_{em} = \frac{p}{2}\lambda_{PM}\left(i_d, i_q\right)i_q \quad (11)$$

This relationship indicates that the primary mechanism of torque production in an SPM machine arises from the interaction between the stator q-axis current and the magnetic field generated by the permanent magnets.

3.2. IPM Modeling

As shown in Section 2.2, an IPM motor uses a more complex rotor structure in which permanent magnets are embedded within the rotor core rather than mounted on the rotor surface. This embedded configuration improves the motor's mechanical robustness, particularly in the rotor, and allows the machine to operate at higher rotational speeds than surface-mounted designs. In addition, the interior placement of the magnets introduces anisotropy in the rotor magnetic circuit, resulting in unequal inductances along the d- and q-axes $\left(L_d \neq L_q\right)$. This magnetic saliency enables the IPM motor to produce torque from both permanent-magnet torque and reluctance torque components. Consequently, the electromagnetic torque of an IPM machine can be expressed as

$$T_{em} = \frac{p}{2}\left[\lambda_{PM} i_q + \left(L_d - L_q\right) i_d i_q\right] \quad (12)$$

The first term represents the magnetic torque generated by the interaction between the stator current and the permanent magnet field, while the second term corresponds to the reluctance torque produced by the difference between the d- and q-axis inductances.

3.3. SynRM Modeling

Unlike permanent magnet machines, the SynRM does not contain permanent magnets in the rotor. Instead, torque is generated by the magnetic saliency created by the rotor geometry. The flux barrier structure produces a significant difference in magnetic reluctance along the d- and q-axis, resulting in unequal inductances along these two axes $\left(L_d \neq L_q\right)$. This anisotropic magnetic characteristic is the fundamental mechanism responsible for torque production in synchronous reluctance machines.

Because no permanent magnet flux exists in the rotor, torque production in a SynRM is purely due to reluctance torque, and can be expressed as

$$T_{em} = \frac{p}{2}\left[\left(L_d - L_q\right) i_d i_q\right] \quad (13)$$

This relationship indicates that torque is produced by the interaction between the stator currents and the rotor's magnetic saliency. Consequently, increasing the difference between the d- and q-axis inductances enhances the machine's reluctance torque capability and improves its overall torque density [34].

3.4. PMASynRM Modeling

As shown in Section 2.3, a PMASyncRM combines the operating principles of synchronous reluctance machines and permanent magnet machines. The rotor typically consists of multiple flux barriers similar to those in a conventional SynRM, with permanent magnets embedded within barriers. These magnets assist the rotor flux distribution while maintaining the saliency characteristics of the reluctance rotor. As a result, they can produce torque through both magnetic and reluctance mechanisms.

Because the PMASyncRM contains both permanent magnet flux and rotor saliency, its electromagnetic torque expression takes the same form as that of the IPM machine described in (12) when the mutual inductances are neglected. Therefore, the total torque consists of the interaction between the stator current and the permanent magnet flux, as well as the reluctance torque generated by the difference between the d- and q-axis inductances. This hybrid torque production mechanism enables improved torque capability compared with conventional synchronous reluctance machines while maintaining strong saliency characteristics in the rotor magnetic circuit [35].

4. Magnet Reduction Methodology for Performance Tradeoff Evaluation

The electromagnetic performance of permanent magnet machines is strongly influenced by the amount and geometry of the permanent magnet material used in the rotor. Variations in magnet dimensions directly affect the machine's electromagnetic torque capability and overall efficiency. Previous studies have shown that permanent magnet volume and its placement in the rotor play a critical role in determining the electromagnetic behavior of traction motors used in electric vehicle applications [36–38]. Therefore, analyzing the impact of reducing magnet material across different machine topologies is important for evaluating the performance trade-offs associated with reduced rare-earth usage in permanent magnet machines.

The magnet geometry can be described using several parameters, including magnet width w_m, magnet length l_m, and radial thickness h_m. Consequently, reducing either the magnet width or magnet length decreases the effective permanent magnet volume and, therefore, the air-gap flux contribution. In machine topologies where torque production depends predominantly on permanent magnet excitation, such as SPM machines, reducing magnet volume directly weakens the air-gap magnetic field and reduces electromagnetic torque capability. In contrast, machine topologies that also exploit rotor saliency, such as IPM and PMASynRM machines, exhibit more complex performance variations because magnet reduction affects not only the permanent magnet torque component but also the interaction between magnetic and reluctance torque production mechanisms. As a result, magnet volume can significantly affect flux linkage, torque density, efficiency, power factor, and flux-weakening capability. However, the extent to which magnet reduction affects overall machine performance across different machine topologies remains insufficiently understood and requires detailed investigation.

Based on these considerations, this section presents a systematic methodology for reducing magnet volume and for comprehensively evaluating the performance trade-offs associated with such reductions in the following sections. Two key geometric parameters, namely magnet width and length, are progressively varied to control the magnet volume while preserving the remaining machine design parameters. This enables the independent quantification of the effects of magnet volume reduction on machine behavior across different topologies. For all machine configurations considered in this study, including SPM, IPM, PMASynRM, and SynRM, we evaluate electromagnetic torque capability, efficiency, flux-linkage characteristics, power factor, and flux-weakening performance under identical stator geometry, winding configuration, material properties, operating conditions, and analysis procedures. Maintaining a consistent evaluation framework ensures that observed performance variations can be attributed primarily to differences in rotor topology and permanent magnet utilization, thereby enabling a fair and unbiased comparison of the performance trade-offs associated with magnet volume reduction.

For all three PM-based topologies, the baseline PM mass is 1.75 kg. Accordingly, the 100%, 80%, 60%, 40%, and 30% PM reduction levels correspond to PM masses of 1.75, 1.40,

1.05, 0.70, and 0.525 kg, respectively. At each percentage level, identical PM mass is maintained across the SPM, IPM, and PMASynRM topologies and between the corresponding magnet-width and magnet-length reduction cases.

The overall methodology adopted for the comparative analysis is summarized in Figure 5. The workflow establishes common geometric and electrical constraints for the investigated machine topologies, defines the topology-specific rotor configurations and PM-reduction scenarios, and subsequently applies the FEA-based electromagnetic analysis. The resulting operating-point data are then used to evaluate and compare the torque–speed and efficiency–speed characteristics of the investigated configurations.

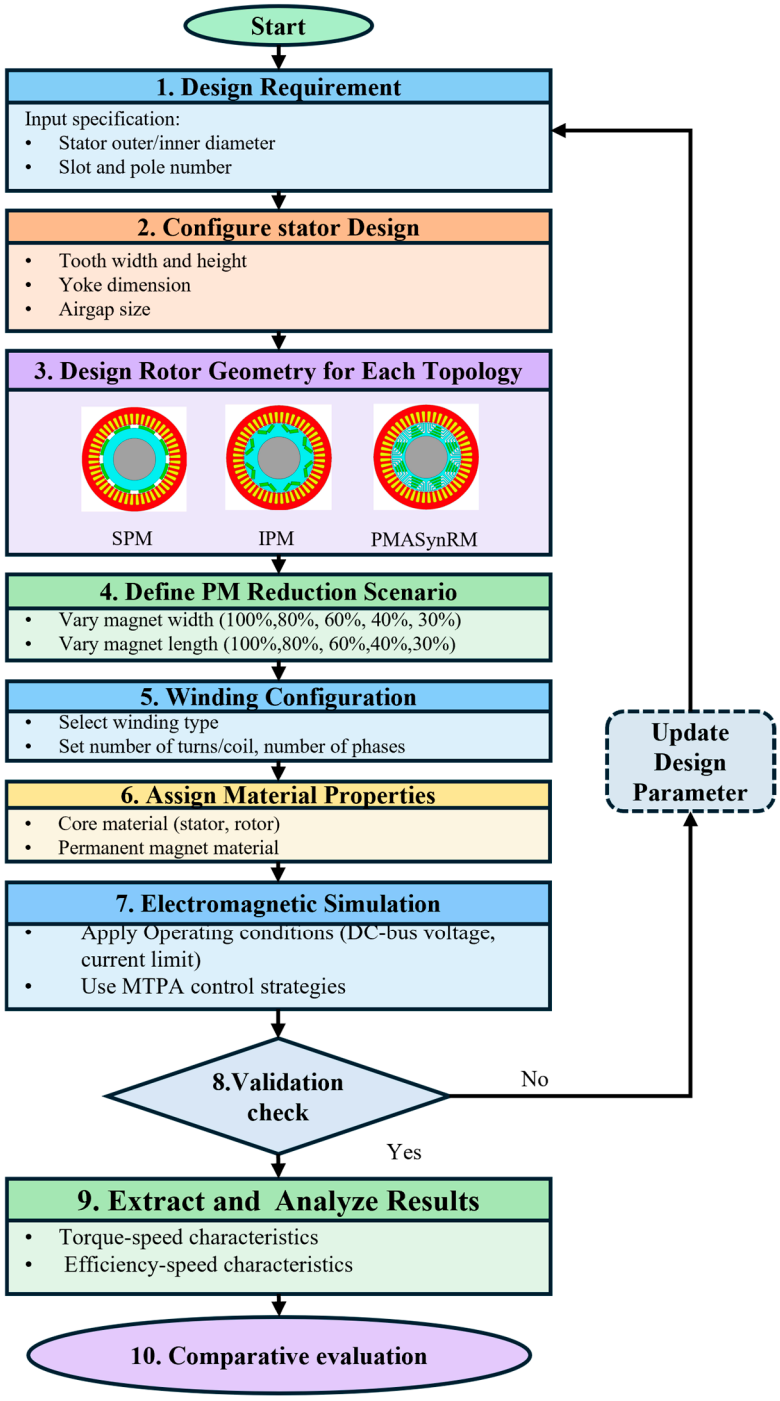


Figure 5. Overall workflow of the proposed methodology for PM-reduction analysis and comparative performance evaluation of SPM, IPM, and PMASynRM machines.

4.1. PM Reduction Scenario in SPM

To investigate how permanent magnet volume affects SPM performance, the magnet dimensions of the SPM machine are systematically varied. In this study, the permanent magnet volume is progressively reduced by modifying the magnet width and length, respectively, while keeping all other machine parameters unchanged. Figure 6 defines the magnet width and length in the SPM machine and shows two representative approaches for reducing the magnet, as adopted in this paper. The magnet reduction is implemented in discrete levels corresponding to 100%, 80%, 60%, 40%, and 30% of the original magnet dimensions. This methodology enables evaluation of the individual influence of magnet width and length on the machine's electromagnetic behavior and overall performance.

For each magnet reduction level, the SPM machine is analyzed over a wide operating speed range to assess how permanent magnet volume affects its electromagnetic behavior. Because torque production in SPM machines is primarily determined by permanent magnet excitation, reducing the magnet dimensions directly weakens the air-gap magnetic field and consequently decreases the machine's torque-producing capability. The resulting performance changes are then compared with those of the other machine topologies.

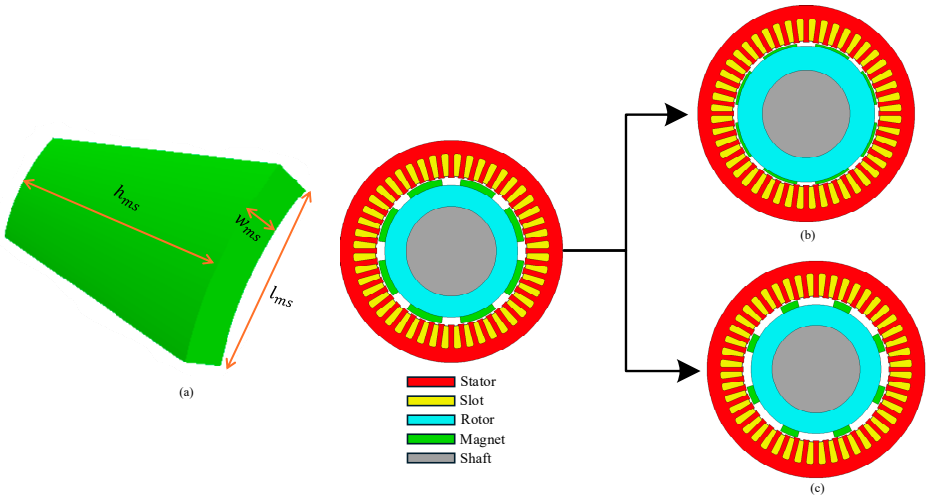


Figure 6. PM reduction scenario in SPM. (**a**) PM Geometric Parameterization, (**b**) a width reduction illustration, and (**c**) a length reduction illustration.

4.2. PM Reduction Scenario in IPM

A similar magnet-reduction methodology is applied to the IPM motor, in which the permanent magnet volume is systematically reduced by independently varying the magnet width and length while keeping the rotor geometry, stator configuration, and operating conditions unchanged. Figure 7 illustrates the definitions of the magnet width and length in the IPM machine, along with two representative IPM magnet-reduction approaches. The reduction is implemented in discrete steps corresponding to 100%, 80%, 60%, 40%, and 30% of the original magnet dimensions. This methodology enables a systematic evaluation of the independent effects of reductions in magnet width and length on the electromagnetic performance of the IPM machine.

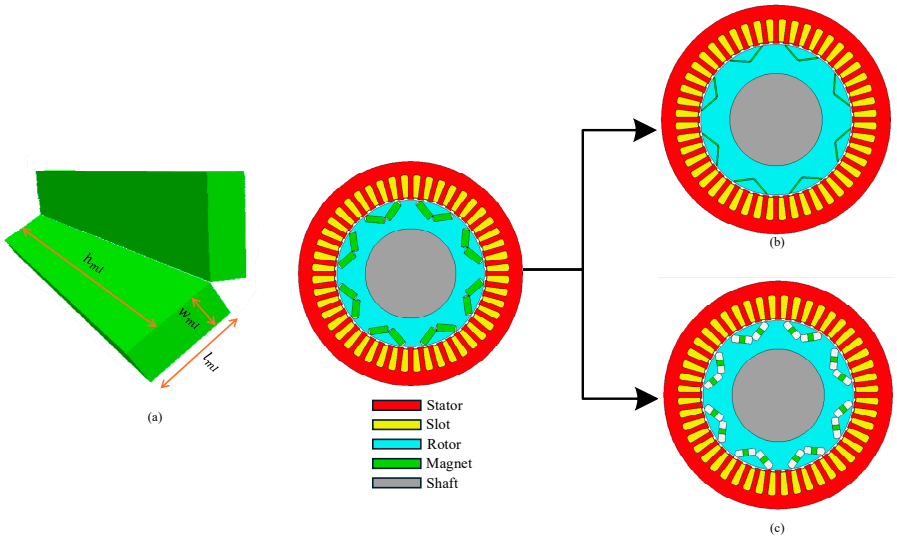


Figure 7. PM reduction scenario in IPM. (**a**) PM Geometric Parameterization, (**b**) a width reduction illustration, and (**c**) a length reduction illustration.

Unlike SPM machines, IPM machines produce torque from both permanent magnet and reluctance components due to rotor magnetic saliency. Therefore, reducing the permanent magnet dimensions not only affects the permanent-magnet flux contribution but may also influence the machine's overall electromagnetic behavior through the interaction between magnetic and reluctance torque components. As a result, the performance sensitivity of IPM machines to magnet reduction may differ from that of SPM machines.

4.3. PM Reduction Scenario in PMASynRM

An analogous magnet-reduction methodology is applied to the PMASynRM machine. As in the SPM and IPM cases, the permanent magnet volume is systematically reduced by varying the magnet width and length while keeping stator geometry, rotor topology, and operating conditions identical. Figure 8 defines the magnet width and length for the PMASynRM machine and shows two representative PMASynRM magnet reduction approaches. The reduction is implemented in discrete steps corresponding to 100%, 80%, 60%, 40%, and 30% of the original magnet dimensions. This approach isolates the effect of permanent magnet volume reduction, enabling consistent evaluation and direct comparison of resulting changes in electromagnetic performance across different machine topologies.

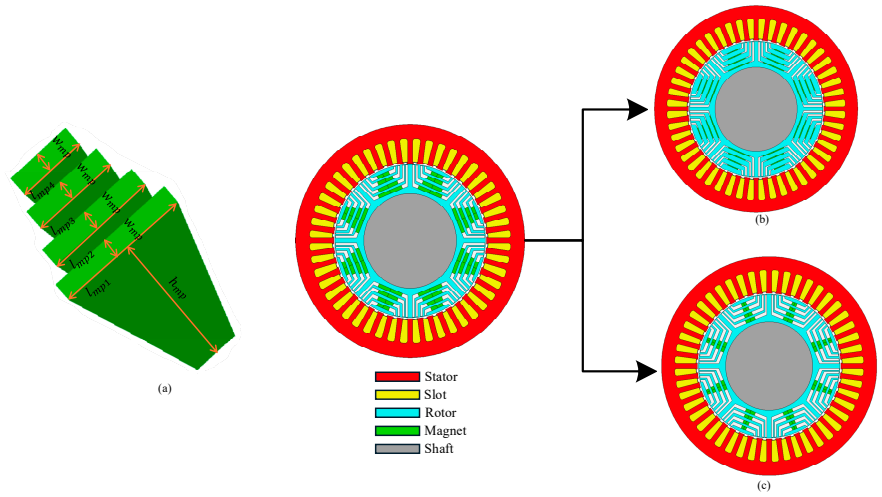


Figure 8. PM reduction scenario in PMASynRM. (**a**) PM Geometric Parameterization, (**b**) a width reduction illustration, and (**c**) a length reduction illustration.

In contrast to SPM machines, where torque production is primarily dependent on permanent magnet excitation, the PMASynRM produces torque through a combination of reluctance torque and magnet-assisted torque, enabled by rotor saliency and permanent magnet flux. Therefore, reducing the magnet dimensions mainly affects the magnet-assisted torque component, while the reluctance torque component associated with rotor saliency remains. As a result, the PMASynRM is expected to show a different sensitivity to magnet reduction than conventional permanent magnet machines.

5. Comparative Performance Evaluation Under Permanent Magnet Volume Reduction

This section presents a comprehensive evaluation and comparative analysis of the electromagnetic performance of SPM, IPM, PMASynRM, and SynRM machine topologies under various permanent magnet reduction conditions. Using the finite element analysis (FEA) models developed in the previous sections, the influence of magnet volume variation is systematically investigated by independently varying magnet width and length while keeping the stator geometry, winding configuration, material properties, and operating conditions identical. Key performance indicators, including electromagnetic torque capability, efficiency, flux linkage characteristics, torque density, and high-speed operating performance, are analyzed over a wide operating speed range. Through this comparative assessment, the impact of permanent magnet reduction on torque production mechanisms, saliency utilization, and overall machine performance is quantified, providing insight into the trade-offs among magnet usage, machine efficiency, and operational capability across different machine topologies.

5.1. FEA Efficiency Map and Extraction of Performance Characteristics

Figure 9 presents the FEA-derived efficiency map of the baseline IPM machine with 100% permanent-magnet volume as a representative example of the performance maps generated in this study. The map illustrates the simulated operating region in terms of electromagnetic torque and rotational speed, with the corresponding efficiency represented by the contour levels. It provides a comprehensive representation of machine performance over the investigated torque–speed operating range and shows the variation in efficiency under different combinations of torque and speed.

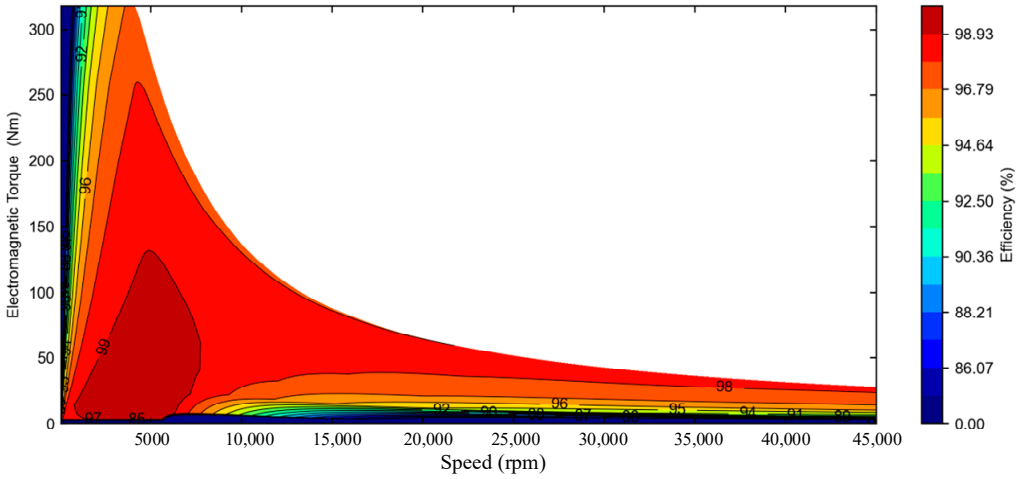


Figure 9. FEA-derived torque–speed efficiency map of the baseline IPM machine with 100% permanent-magnet volume.

Similar FEA performance maps were generated for the investigated SPM, IPM, and PMASynRM configurations at the different PM reduction levels. Although FEA simulations provide valuable information on the performance of PMSMs, direct comparison of FEA-derived efficiency maps alone does not provide sufficient quantitative insight for a comprehensive comparative study. Therefore, custom-developed data-processing programs were developed to systematically extract key performance characteristics and quantitative metrics. These extracted parameters enable consistent and objective comparisons among different machine topologies and permanent magnet reduction conditions, providing a more informative assessment of their efficiency and overall performance across the operating speed and torque ranges.

The torque–speed and efficiency–speed characteristics reported throughout this study, which represent the maximum torque and corresponding efficiency boundaries, were extracted from the underlying operating-point data used to generate these FEA efficiency maps. For each machine configuration and PM reduction level, the corresponding electromagnetic torque and average efficiency values were obtained at the investigated operating speeds and used to construct the maximum torque–speed and efficiency–speed boundary curves presented in the subsequent sections. Therefore, the performance curves presented in Figures 10–15 are derived directly from the underlying FEA operating-point data. Figure 9 is presented as a representative example to illustrate this data-extraction procedure and to provide additional verification and transparency regarding the generation of the reported performance characteristics.

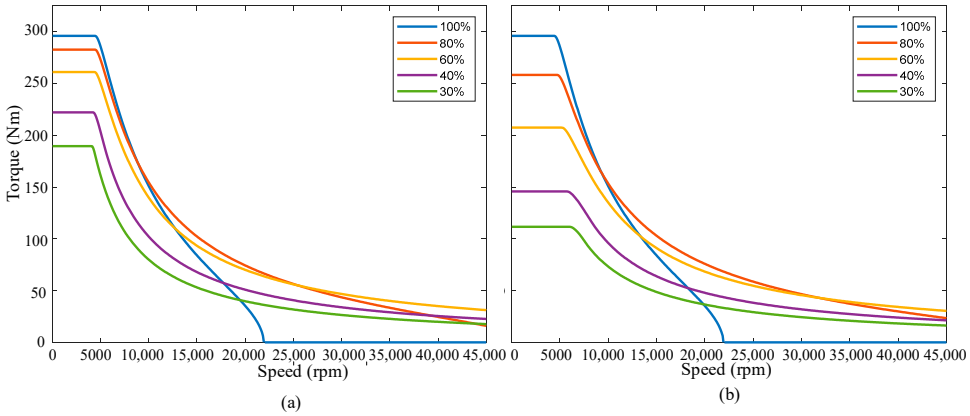


Figure 10. Torque–speed characteristics of the SPM machine under magnet dimension variations: (**a**) effect of magnet width reduction and (**b**) effect of magnet length reduction.

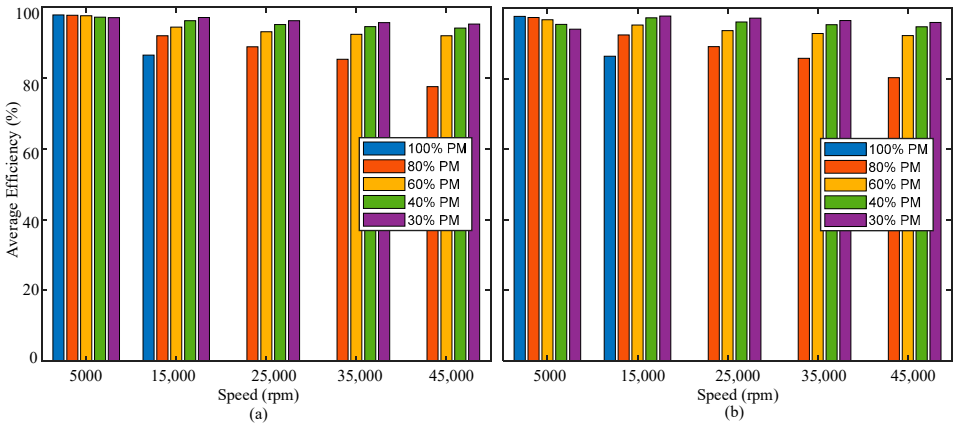


Figure 11. Average Efficiency comparison of the SPM machine at different operating speeds for various permanent magnet reduction levels. Results are shown for (**a**) magnet-width reduction and (**b**)

magnet-length reduction, where the permanent magnet dimensions are reduced from 100% to 30% of their original values.

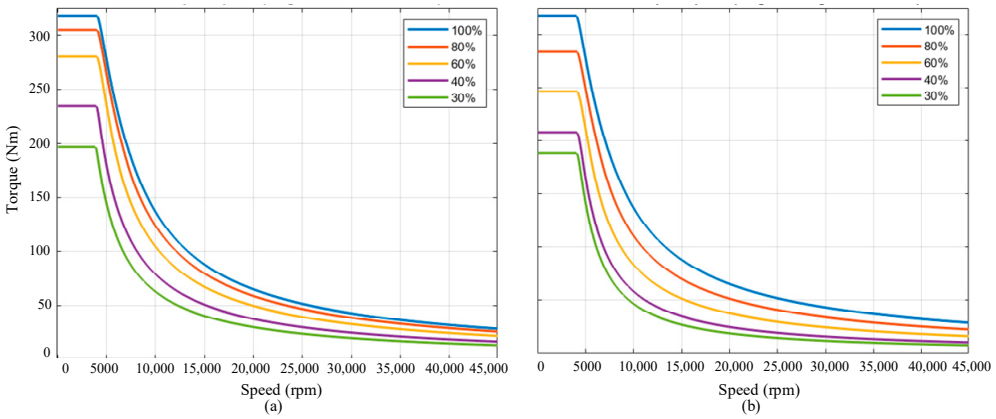


Figure 12. Torque–speed characteristics of the IPM machine under magnet dimension variations: (**a**) effect of magnet width reduction and (**b**) effect of magnet length reduction.

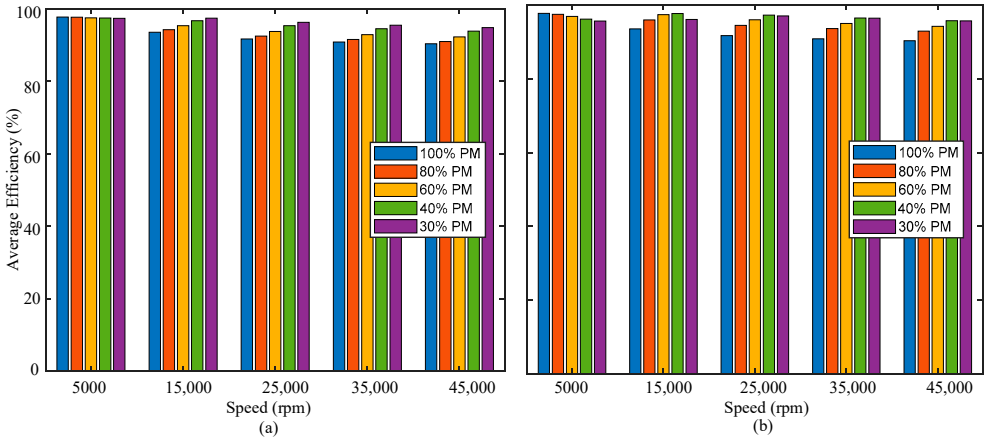


Figure 13. Average Efficiency comparison of the IPM machine at different operating speeds for various permanent magnet reduction levels. Results are shown for (**a**) magnet-width reduction and (**b**) magnet-length reduction, where the permanent magnet dimensions are reduced from 100% to 30% of their original values.

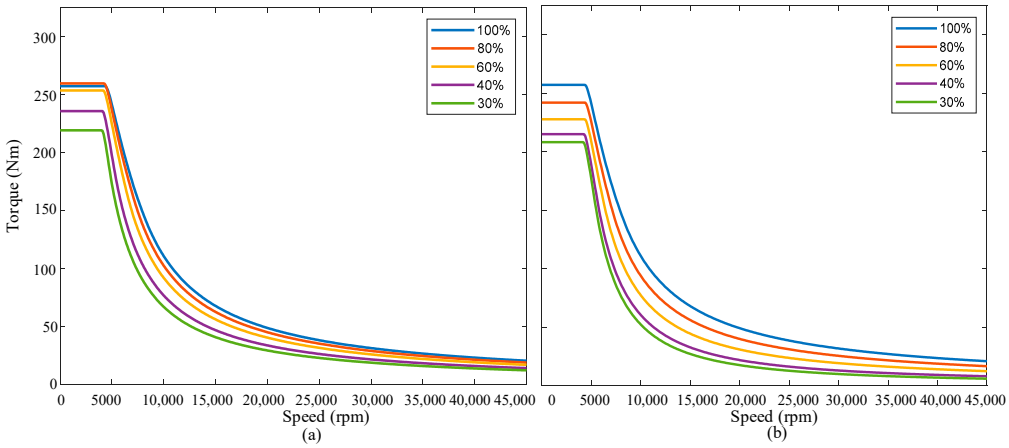


Figure 14. Torque–speed characteristics of the PMASynRM machine under magnet dimension variations: (**a**) effect of magnet width reduction and (**b**) effect of magnet length reduction.

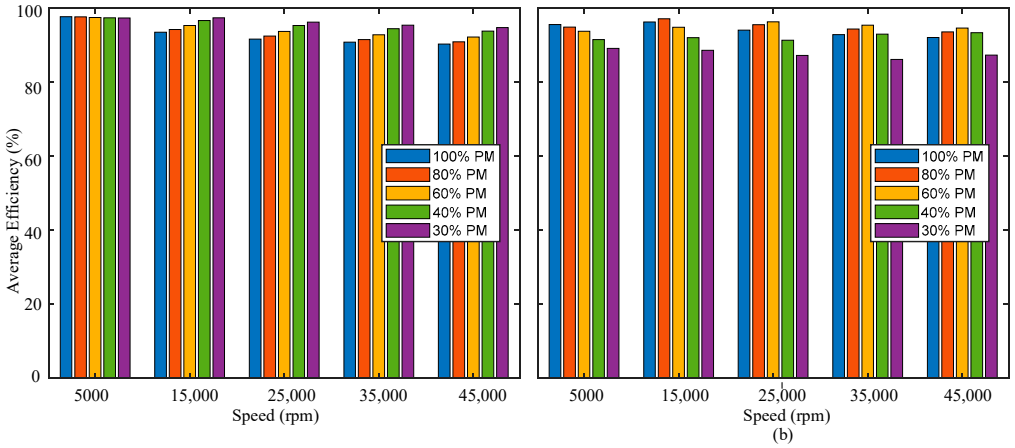


Figure 15. Average Efficiency comparison of the PMASynRM machine at different operating speeds for various permanent magnet reduction levels. Results are shown for (**a**) magnet-width reduction and (**b**) magnet-length reduction, where the permanent magnet dimensions are reduced from 100% to 30% of their original values.

5.2. Evaluation of SPM with Varying Permanent Magnet Volume

Figure 10 presents the maximum torque-speed contour characteristics [3] of the SPM machine under progressive reduction in magnet width and length. As shown in the figure, reducing the permanent magnet dimensions decreases the available air-gap flux and consequently lowers the electromagnetic torque capability across the operating range. The effect is most pronounced in the constant-torque region, where torque production is primarily governed by permanent magnet excitation. As the PM dimensions decrease from 100% to 30% of the baseline design, a gradual reduction in peak torque is observed across both reduction strategies. Furthermore, degradation is more severe with a reduction in magnet length than with a reduction in magnet width, indicating that the machine is more sensitive to variations in magnet length. This behavior can be attributed to the greater reduction in effective magnetic flux linkage caused by the shortened magnetization path, which directly weakens the air-gap field responsible for torque production.

To further investigate the relative impact of the two reduction strategies, a direct comparison of the torque–speed characteristics shown in Figure 10a,b is conducted. The comparison demonstrates that reducing magnet width preserves higher torque capability than reducing magnet length across the entire speed range. The difference is relatively small at the 80% PM level but becomes increasingly pronounced as the PM dimensions are reduced to 60%, 40%, and 30% of their original values. This trend suggests that reducing magnet width allows more of the original magnetic circuit to remain effective, whereas reducing magnet length more directly weakens the magnet's flux linkage and electromagnetic loading. Consequently, from a torque-production perspective, reducing magnet width offers a more favorable trade-off between permanent magnet material savings and machine performance. Figure 10b also reveals an important high-speed operating characteristic of the SPM machine. At low operating speeds, particularly below base speed, a larger permanent magnet volume yields higher available torque because of increased air-gap flux and stronger permanent magnet excitation. However, at higher operating speeds, increasing magnet volume can significantly reduce achievable torque. This behavior is primarily caused by the stator voltage limitation imposed by the PWM saturation constraint of the motor inverter, as described by the flux linkage constraint in (8). As the motor speed increases, the back electromotive force and stator voltage demand rise proportionally with the air-gap flux (5). To satisfy the inverter voltage limit, flux weakening becomes necessary. Under flux-weakening operation, a larger negative d-axis current (i_d) is injected

to counteract the permanent magnet flux, thereby reducing the available q-axis current (i_q) responsible for torque production.

According to the electromagnetic torque relation in (9), this current redistribution reduces the achievable torque output at high speeds. As motor speed increases, the magnitude of (i_d) approaches the rated current limit, while (i_q) progressively decreases toward zero due to the stator current constraint defined in (5). Consequently, although larger permanent magnet volumes provide superior low-speed torque capability, they can adversely affect high-speed torque performance by increasing the flux-weakening requirement. This behavior represents an important operating characteristic of SPM machines and highlights the trade-off between low-speed torque enhancement and high-speed operating capability.

The influence of PM reduction on the flux-weakening region can be further explained through the relationship among PM flux linkage, back-EMF, and the inverter voltage constraint. As the PM dimensions are reduced, the PM-generated flux linkage decreases, which consequently reduces the back-EMF developed at a given rotational speed. Therefore, the stator-voltage limit is reached under different operating conditions, and a lower magnitude of negative d-axis current may be required to counteract the PM flux during high-speed operation. This reduction in PM excitation can facilitate extension of the usable high-speed operating region under the imposed inverter voltage constraint.

However, this extension is accompanied by a reduction in electromagnetic torque capability. Although lower PM excitation alleviates the high-speed back-EMF constraint, it also decreases the PM contribution to torque production. Consequently, the reduced-PM configurations exhibit a trade-off between an extended high-speed operating region and reduced torque capability. The results in Figure 10 therefore demonstrate that PM reduction influences not only the magnitude of electromagnetic torque but also the onset and extent of flux-weakening operation. Furthermore, the differences between magnet-width and magnet-length reduction indicate that the direction of PM dimensional reduction also affects this trade-off.

While Figure 10 demonstrates the influence of PM reduction on the torque–speed and flux-weakening characteristics of the SPM machine, Figure 11 presents the corresponding average-efficiency behavior at selected operating speeds. At lower speeds, the efficiency differences among the investigated PM levels remain relatively small because the machine operates with less severe voltage limitation and the available stator current can be used predominantly for torque production. As the machine enters the flux-weakening region, however, the efficiency characteristics become increasingly dependent on the PM excitation level and the corresponding allocation of the available stator current.

In the flux-weakening region, part of the available stator current must be allocated to negative i_d to weaken the effective PM flux and maintain the stator voltage within the inverter limit. Consequently, the current vector is no longer determined solely by the requirement for torque production i_q; instead, it must satisfy both flux-weakening and torque-production requirements within the fixed current constraint. As established in the preceding torque–speed analysis, stronger PM excitation requires greater flux weakening at elevated speeds. This changes the distribution of the available current between the d and q-axis components and increasingly restricts the feasible electromagnetic operating region. The resulting current utilization helps explain why the efficiency differences among the PM configurations become more pronounced in the high-speed region.

Reducing the PM dimensions changes this balance. With lower PM excitation, the machine requires less opposing d-axis current to remain within the voltage-limited operating region. A larger portion of the available current capability can therefore remain associated with the torque-producing component, reducing the severity of the current re-

distribution required for flux weakening. Accordingly, the reduced-PM configurations exhibit better efficiency retention as the operating speed increases. It indicates that an appropriate reduction in PM excitation can improve the utilization of the available voltage and current capability once the SPM machine enters extended flux-weakening operation.

However, reducing the PM volume also decreases the PM torque contribution. Therefore, maintaining the same torque demand requires a higher stator current, which can increase motor losses and partially offset the efficiency benefit obtained from reduced flux-weakening requirements. These competing effects explain why the efficiency does not increase proportionally with PM reduction, as observed in Figure 11. The overall efficiency benefit of PM reduction is therefore determined by the balance between the reduced current required for flux weakening and the increased current required to compensate for the loss of PM torque. This trade-off highlights the existence of an optimal PM volume at which the benefits of reduced flux-weakening demand can be achieved without incurring excessive additional current and associated losses.

The magnet-width and magnet-length reduction cases exhibit broadly similar efficiency trends because both strategies reduce PM excitation and consequently alter the current requirement during flux weakening. The differences between the two strategies are relatively modest from an efficiency perspective, with magnet-length reduction showing a slightly higher efficiency at several high-speed operating points. However, the torque–speed results in Figure 10 show a substantially stronger distinction between the two reduction directions, with magnet-width reduction preserving greater torque capability. Therefore, the efficiency results should be interpreted together with the torque characteristics: PM reduction alleviates the electrical constraints associated with extended flux weakening, but the geometric direction of reduction determines how effectively torque capability is retained while obtaining this high-speed benefit. For the investigated SPM configuration, magnet-width reduction consequently provides the more favorable overall compromise between torque retention, current utilization, efficiency retention, and extended high-speed operation.

5.3. Evaluation of IPM with Varying Permanent Magnet Volume

Figure 13 shows the torque-speed characteristics of the IPM machine under progressive reduction in magnet width and length. As shown, gradually reducing the permanent magnet dimensions decreases electromagnetic torque across the operating speed range. Similar to the SPM machine, the reduction is most pronounced in the constant-torque region, where permanent magnet excitation contributes significantly to torque production. However, unlike the SPM topology, the IPM machine continues to generate reluctance torque due to its rotor saliency. Consequently, although permanent magnet torque decreases as magnet volume decreases, the reluctance torque component partially compensates for this reduction, leading to a more gradual degradation in overall torque capability. As the PM dimensions decrease from 100% to 30% of the baseline design, both magnet reduction strategies yield lower peak torque, although the degradation remains less severe than would be expected from the reduction in magnet volume alone.

To further evaluate the influence of the two reduction strategies, a direct comparison of the torque-speed characteristics shown in Figure 12a,b is performed. The comparison indicates that magnet width reduction consistently preserves higher torque capability than magnet length reduction across the entire operating speed range. As with the SPM machine, the difference is relatively small at the 80% PM level but grows as the magnet dimensions are reduced to 60%, 40%, and 30% of their original values. This behavior can be explained by the fact that reducing the magnet width maintains a larger effective magnetization path and preserves more air-gap flux, while reducing the magnet length more

directly weakens the permanent magnet excitation. Although reluctance torque from rotor saliency still contributes to the total torque, it cannot fully compensate for the larger reduction in permanent magnet torque associated with magnet length reduction. Therefore, from a torque-production perspective, magnet width reduction offers a better compromise between permanent magnet material savings and electromagnetic performance in the IPM machine.

Unlike the SPM machine, the IPM machine exhibits improved high-speed operating characteristics due to the presence of rotor saliency. At low operating speeds, larger permanent magnet volumes produce higher torque due to stronger permanent magnet excitation. As the operating speed increases, however, the reluctance torque component becomes increasingly significant, enabling the machine to maintain higher torque during flux-weakening operation. According to Equation (10), the electromagnetic torque comprises both permanent magnet and reluctance torque components. During high-speed operation, the inverter injects negative d-axis current to satisfy the voltage constraint Equation (8). While this current reduces the permanent magnet flux linkage, it simultaneously increases the reluctance torque via the saliency term $(L_q - L_d)i_d i_q$. Consequently, the IPM machine experiences less degradation in high-speed torque capability than the SPM machine, demonstrating the inherent advantage of rotor saliency under voltage-limited operating conditions.

While Figure 12 demonstrates the influence of PM reduction on the torque–speed characteristics of the IPM machine, Figure 13 presents the corresponding average-efficiency behavior at selected operating speeds. Similar to the SPM machine, in the lower-speed region, the efficiency differences among the investigated PM configurations remain relatively small. As the operating speed increases and the machine moves further into the flux-weakening region, however, the effect of PM reduction becomes more evident. Compared to the SPM machine, all investigated IPM configurations maintain relatively high efficiency throughout the extended operating-speed range, demonstrating that substantial changes in PM dimensions can be accommodated without a severe deterioration in efficiency.

At high speeds, the efficiency trend is closely associated with the current utilization and flux-weakening behavior established in the preceding torque–speed analysis. As discussed for Figure 12, the d-axis current required for flux weakening in the IPM machine also participates in reluctance-torque production because of rotor saliency. Consequently, the redistribution of the current vector during high-speed operation is more effectively utilized for electromagnetic torque production than in the SPM topology. Reducing PM excitation further alleviates the flux-weakening requirement, allowing the machine to operate within the imposed voltage and current constraints with a more favorable balance between the PM and reluctance contributions. Again, the results for IPM machines also show that the overall efficiency benefit of PM reduction is determined by the balance between the reduced current required for flux weakening and the increased current required to compensate for the loss of PM torque.

For magnet-width reduction (Figure 13a), the efficiency differences among the PM levels are relatively limited at lower speeds but become progressively more apparent as speed increases. Greater magnet-width reduction generally improves efficiency retention in the high-speed region. This trend indicates that the reduction in PM excitation becomes increasingly beneficial once operation is dominated by the voltage constraint and flux-weakening requirement. At the same time, the reluctance-torque contribution allows the machine to partially compensate for the reduced PM excitation. Therefore, the high-speed efficiency behavior reflects the ability of the IPM topology to redistribute electromagnetic loading between PM and reluctance torque as the magnet width is reduced.

For magnet-length reduction (Figure 13b), PM reduction also improves efficiency retention in the intermediate- and high-speed regions, although the trend is not strictly proportional to the degree of reduction. Beyond a certain reduction level, further decreasing the magnet length produces little additional efficiency benefit. This suggests that once the PM excitation has been sufficiently reduced to alleviate the high-speed flux-weakening burden, further PM reduction provides diminishing improvement in the electromagnetic operating condition. The efficiency behavior therefore reflects a balance between reducing the PM-related flux-weakening requirement and retaining sufficient PM excitation to complement the reluctance-torque component.

A comparison of Figure 13a,b shows that both reduction strategies provide favorable high-speed efficiency characteristics, with magnet-length reduction showing a modest efficiency advantage at several corresponding operating conditions. However, this advantage should be considered together with Figure 12, where magnet-width reduction provides better torque retention. Thus, the two reduction strategies influence torque and efficiency differently: magnet-length reduction can provide slightly more favorable high-speed efficiency, whereas magnet-width reduction better preserves electromagnetic torque. Considering the torque–speed and efficiency results together, the IPM topology demonstrates substantial tolerance to PM reduction because its reluctance-torque capability supports effective current utilization during flux weakening. The preferred reduction strategy therefore depends on the desired balance among PM-material reduction, torque retention, flux-weakening operation, and high-speed efficiency.

5.4. Evaluation of PMASynRM with Varying Permanent Magnet Volume

Figure 14 presents the torque-speed characteristics of the PMASynRM machine under progressive reductions in magnet width and length. Similar to the SPM and IPM machines, reducing the permanent magnet dimensions decreases the available permanent magnet flux, lowering electromagnetic torque across the operating speed range. However, torque degradation is less pronounced than in the SPM and IPM machines because the PMASynRM generates torque from both permanent magnet torque and a larger reluctance torque component produced by its highly salient rotor structure. As the permanent magnet dimensions decrease from 100% to 30% of the baseline design, the peak torque gradually decreases for both reduction strategies, while a considerable portion of the torque capability is retained due to the stronger reluctance torque component. This demonstrates that the PMASynRM topology is more robust to permanent magnet volume reduction than conventional permanent magnet machines.

To further examine the influence of the two reduction strategies, a direct comparison of the torque-speed characteristics shown in Figure 14a,b is conducted with IPMs; the results indicate that reducing the magnet width consistently preserves higher torque capability than reducing the magnet length across the entire operating speed range. This behavior indicates that reducing magnet width better preserves the effective permanent magnet flux linkage while maintaining the rotor saliency responsible for producing reluctance torque. In contrast, reducing magnet length reduces permanent magnet excitation more, resulting in a larger decrease in the magnetic torque component. Consequently, reducing magnet width provides a more favorable balance between permanent magnet material reduction and torque capability in the PMASynRM machine.

Compared with the SPM machine, the PMASynRM delivers superior high-speed operating performance due to its strong rotor saliency and reduced reliance on permanent magnet excitation. At low speeds, larger permanent magnet volumes increase torque by raising the air-gap flux and magnetic torque component. As the speed increases, however, the machine gradually enters the flux-weakening region, where the reluctance torque con-

tribution becomes increasingly dominant. Because a significant share of the total electromagnetic torque arises from rotor saliency rather than permanent magnet excitation alone, reducing permanent magnet volume has a comparatively smaller impact on high-speed torque capability.

While Figure 14 illustrates the influence of PM reduction on the torque–speed characteristics of the PMASynRM machine, Figure 15 presents the corresponding average-efficiency behavior at selected operating speeds. At lower speeds, the magnet-width reduction cases exhibit relatively small differences in efficiency, whereas the influence of PM reduction becomes increasingly apparent as the operating speed increases. All investigated configurations remain operational throughout the considered speed range, although the efficiency response differs considerably between magnet-width and magnet-length reduction. This distinction demonstrates that the high-speed efficiency of the PMASynRM depends not only on the amount of PM material but also on how the remaining magnets interact with the reluctance-based torque-production mechanism of the rotor.

For magnet-width reduction (Figure 15a), the efficiency remains comparatively insensitive to PM reduction at lower speeds, while progressively reduced magnet width generally provides better efficiency retention as speed increases. In the flux-weakening region, reduced PM excitation lowers the influence of the magnets on the voltage-limited operating condition, while the highly salient rotor continues to provide a substantial reluctance-torque contribution. Consequently, the machine becomes less dependent on PM excitation for torque production as the magnet width is reduced. The available current can therefore support the combined requirements of flux weakening and reluctance-based torque production without a pronounced deterioration in efficiency. This explains the relatively smooth and consistent efficiency characteristics observed across the magnet-width reduction cases, even under substantial PM reduction.

The magnet-length reduction results (Figure 15b) reveal a distinctly different behavior. Moderate reduction initially provides favorable efficiency characteristics because lower PM excitation alleviates the flux-weakening requirement while sufficient PM assistance remains available to complement the reluctance-torque component. However, this benefit does not continue as the magnet length is reduced more aggressively. At severe reduction levels, the decrease in PM assistance becomes sufficiently large that the reluctance-torque mechanism alone cannot fully compensate for the altered electromagnetic operating condition. The current distribution must then satisfy torque production and flux-weakening requirements with substantially weaker PM assistance, resulting in less favorable efficiency characteristics. This explains the non-monotonic trend observed for magnet-length reduction, where moderate reduction can be beneficial but excessive reduction leads to a pronounced deterioration in efficiency.

A direct comparison of Figure 15a,b therefore demonstrates an important interaction between PM geometry and the reluctance-dominated torque-production mechanism of the PMASynRM. Magnet-width reduction produces a more gradual and consistent efficiency response as PM material is removed, indicating that the remaining magnet configuration continues to provide effective PM assistance while preserving the contribution of rotor saliency. Magnet-length reduction, in contrast, can provide favorable efficiency at moderate reduction levels but becomes increasingly detrimental when the reduction is severe. Therefore, the efficiency results reinforce the torque–speed observations in Figure 14: the amount of PM material alone does not determine PMASynRM performance; the geometric distribution of the remaining PM material determines how effectively PM excitation complements reluctance torque during both normal and flux-weakening operation. For the investigated PMASynRM configuration, magnet-width reduction consequently provides the more robust strategy when substantial PM-material reduction is required,

whereas moderate magnet-length reduction can remain advantageous when efficiency is prioritized within a less aggressive reduction range.

5.5. Comparative Evaluation Across Different PMSMs

Figure 16 compares the torque-speed characteristics of the SPM, IPM, and PMASynRM machines with an identical amount of permanent magnet material. Despite equal PM volume, the three topologies exhibit substantially different torque characteristics, indicating that electromagnetic performance depends not only on the quantity of magnet material but also on magnet placement, rotor magnetic circuit, saliency, and the relative contributions of permanent-magnet and reluctance torque.

In the low-speed constant-torque region, the IPM machine produces the highest torque, approximately 317 N·m, followed by the SPM machine at approximately 296 N·m and the PMASynRM at approximately 257 N·m. The high low-speed torque of the SPM machine can be attributed to the direct use of surface-mounted magnets to establish the air-gap field. Because the magnets are positioned directly adjacent to the air gap and the rotor has low magnetic saliency, torque production is dominated by the interaction between the PM flux linkage and the stator q-axis current (Equation (9)). Consequently, a large proportion of the available magnet flux directly contributes to electromagnetic torque, enabling the SPM machine to achieve relatively high torque density at low speed. Furthermore, the absence of a substantial reluctance torque component makes its torque production strongly dependent on the available PM flux [39,40]

The IPM machine achieves even higher low-speed torque because it combines a strong permanent-magnet torque component with additional reluctance torque arising from rotor saliency. The embedded magnets provide the PM excitation, while the difference between the d- and q-axis inductances enables the rotor to develop reluctance torque under appropriate current excitation. Therefore, under MTPA operation, both torque-producing mechanisms can be used simultaneously, allowing the investigated IPM machine to achieve the highest torque among the three topologies despite having the same PM material volume [40].

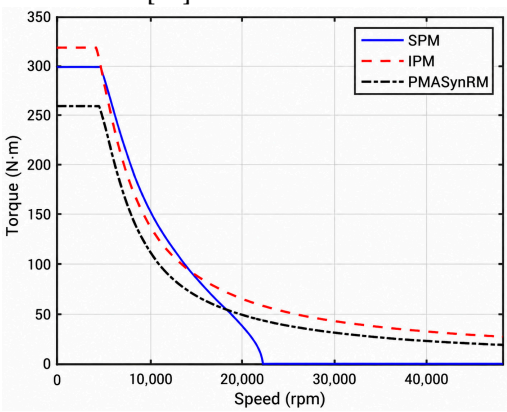


Figure 16. Torque–speed characteristics of the SPM, IPM, and PMASynRM machines with identical permanent magnet amount, illustrating the influence of rotor topology on torque capability and high-speed operating performance.

The lower torque of the investigated PMASynRM should therefore not be interpreted as a consequence of reduced magnet quantity. Rather, it results from the different way the same magnet volume is used within its flux-barrier rotor.

The PMASynRM is derived from a highly salient reluctance rotor, in which multiple flux barriers are primarily designed to establish magnetic anisotropy. The magnets placed inside these barriers assist in producing reluctance torque but do not necessarily produce the same effective air-gap PM flux linkage as magnets located in an SPM or conventional

IPM rotor. A portion of the rotor volume is occupied by flux barriers, and the resulting flux paths are strongly influenced by barrier geometry, iron bridges, magnetic saturation, and leakage flux. Consequently, the same PM volume can yield a lower effective PM torque contribution. Although the PMASynRM's high saliency provides substantial reluctance torque, this contribution is not sufficient in the investigated design to fully compensate for the lower effective PM torque, resulting in a lower total low-speed torque than that of the IPM and SPM machines [41,42].

In contrast, the IPM and PMASynRM machines exploit rotor saliency and reluctance torque at high speed. A negative d-axis current used for flux weakening not only reduces the effective air-gap flux but also contributes to reluctance torque through the saliency term. As a result, these machines can use the available current more effectively under voltage-limited conditions and sustain torque over a substantially wider speed range. Among the investigated designs, the IPM provides the most favorable overall torque-speed characteristic by combining the highest low-speed torque with strong high-speed torque retention. The PMASynRM sacrifices some low-speed torque because it uses magnets primarily to assist a strongly reluctance-oriented rotor. In contrast, the SPM converts the available PM material very effectively into low-speed torque but shows the greatest limitation during extended flux-weakening operation. Therefore, even with identical PM material usage, the three machines demonstrate markedly different trade-offs between low-speed torque density and high-speed operating capability.

Following the torque–speed comparison in Figure 16, Figure 17 compares the average efficiencies of the SPM, IPM, and PMASynRM machines under the 100% PM configuration at selected operating speeds. In the lower-speed region, all three topologies exhibit high average efficiency with relatively small differences among them. This indicates that when operation is less constrained by the inverter voltage limit, the different rotor structures can all utilize the available PM excitation effectively. As speed increases, however, the efficiency characteristics begin to diverge, demonstrating the increasing influence of rotor topology, saliency, and the corresponding flux-weakening requirements.

The SPM machine exhibits a more restricted high-speed operating range under the baseline PM configuration. Because its torque production depends predominantly on PM excitation and its reluctance-torque contribution is comparatively small, the increasing voltage demand at elevated speed requires a substantial portion of the available current capability to be devoted to flux weakening. As the machine moves further into the voltage-limited region, this current redistribution increasingly restricts the feasible operating condition. Consequently, although the SPM topology provides high efficiency and strong torque performance at lower speeds, its 100% PM configuration cannot sustain the extended high-speed operation achieved by the other two topologies.

In contrast, both the IPM and PMASynRM machines maintain high average efficiency throughout the extended operating-speed range. Their greater rotor saliency enables reluctance torque to complement PM torque as the machines enter flux-weakening operation. Thus, the negative d-axis current required for flux weakening also participates in reluctance-torque production rather than serving only to oppose PM excitation. This provides more effective utilization of the available current under voltage-limited conditions and reduces the dependence of torque production on PM excitation alone. The PMASynRM particularly benefits from its reluctance-dominated torque-production mechanism, while the IPM maintains a stronger combination of PM and reluctance torque.

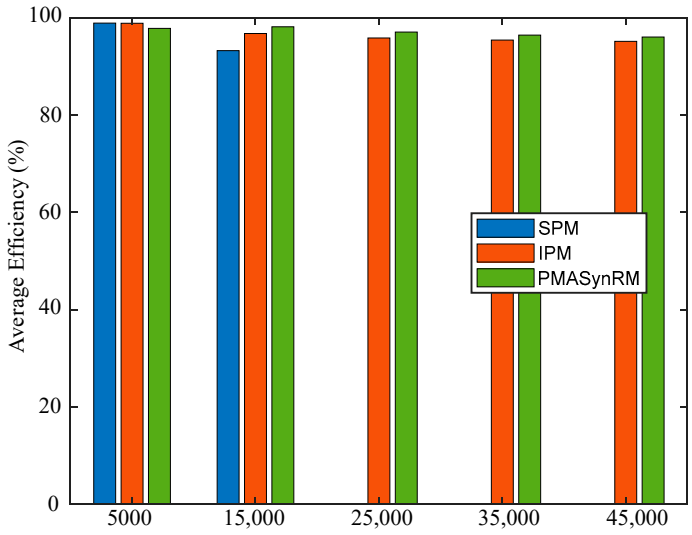


Figure 17. Average Efficiency comparison of SPM, IPM, and PMaSynRM machines under the 100% PM configuration across different operating speeds.

Considering both the torque–speed and average-efficiency characteristics, the results demonstrate that identical PM material usage does not result in equivalent performance across the investigated machine topologies. The IPM configuration provides the highest baseline torque and maintains high average efficiency over the extended operating-speed range, reflecting the combined contribution of PM and reluctance torque. The SPM configuration exhibits strong low-speed performance and high average efficiency but has a more limited high-speed operating range under the 100% PM condition. In comparison, the PMASynRM produces lower baseline torque but maintains a broad operating-speed range and competitive average efficiency, demonstrating the contribution of rotor saliency and reluctance torque to reduced-PM operation. Furthermore, the PM-reduction results show that both the amount and geometric direction of magnet reduction influence machine performance, with the magnitude of these effects varying among the three topologies. These findings demonstrate that effective PM utilization depends not only on the quantity of magnetic material but also on its geometric distribution and interaction with the topology-specific rotor structure, highlighting the importance of considering torque capability, efficiency, high-speed operation, and PM usage together when evaluating reduced-magnet traction-machine configurations.

6. Conclusions

This study presents a unified FEA-based methodology for systematically evaluating permanent-magnet (PM) reduction across SPM, IPM, and PMASynRM machines under consistent design and operating conditions. The methodological contribution lies in independently varying magnet width and magnet length from 100% to 30% of their baseline dimensions while maintaining common machine dimensions, winding configurations, material properties, and operating constraints. This controlled framework enables direct quantification of topology-dependent sensitivity to both the amount and direction of PM dimensional reduction. The comparative results demonstrate that magnet-width reduction generally preserves torque and efficiency more effectively than magnet-length reduction, with the difference becoming increasingly pronounced at lower PM levels.

The results also reveal distinct sensitivities to PM reduction among the investigated topologies. The SPM exhibits the greatest sensitivity because its torque production predominantly depends on PM excitation. In contrast, the IPM and PMASynRM better tolerate PM reduction because rotor saliency and the associated reluctance torque partially

compensate for the reduced PM contribution and support extended high-speed operation. The equal-PM comparison further demonstrates that identical magnet quantity does not yield equivalent electromagnetic performance. With the same PM amount, the IPM achieves the highest low-speed torque of approximately 317 N·m, compared with 296 N·m for the SPM and 257 N·m for the PMASynRM. The IPM also provides the most favorable overall balance of low-speed torque, high-speed torque retention, and efficiency, while the PMASynRM maintains a broad operating range and relatively high efficiency through greater utilization of reluctance torque. Table 2 provides a summary comparison of the study conducted in this paper.

Table 2. Comparison of the investigated machine topologies under PM reduction.

Comparison Criterion	SPM	IPM	PMASynRM
Primary Torque Production	Predominantly PM torque	PM + reluctance torque	Strong reluctance torque + PM assistance
Rotor PM Placement	Surface-mounted	Interior	Embedded within flux-barrier structure
Rotor Saliency	Low	High	High
Dependence on PM Excitation	High	Moderate	Lower
Reluctance-Torque Contribution	Low	High	High
Sensitivity to PM Reduction	High	Moderate	Lower
Torque Retention with PM Reduction	Lower	High	High
Flux-Weakening Capability	More dependent on PM flux reduction	Strong	Strong
Effect of PM Width Reduction	Better performance retention than length reduction	Better performance retention than length reduction	Better performance retention than length reduction
Effect of PM Length Reduction	Greater performance degradation	Greater degradation at substantial reduction	Greater degradation at substantial reduction
Tolerance to Large PM Reduction	Lower	High	High
Main Trade-off under PM Reduction	Torque capability decreases more noticeably	PM reduction trades PM torque against reluctance contribution	Severe reduction eventually reduces PM assistance and performance
Overall PM-Reduction Behavior Observed in This Study	Most sensitive to PM reduction	Relatively robust	Relatively robust, particularly for width reduction

Overall, the findings demonstrate that reducing PM usage does not necessarily result in a proportional decline in machine performance. Careful selection of rotor topology and PM-reduction geometry can substantially mitigate torque and efficiency penalties. In particular, magnet-width reduction combined with salient rotor topologies, such as IPM and PMASynRM, provides a promising approach for reducing rare-earth magnet consumption while maintaining competitive electromagnetic performance. Future work will consider rotor optimization, thermal and mechanical constraints, demagnetization risk, drive-cycle performance, and experimental validation.

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